Zihang Wu

Algorithms, Optimization & Data Structures
PhD Researcher, Max Planck Institute for Informatics
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Profile

Computer science researcher with papers at FOCS, SoCG, and SODA. Combines theoretical algorithm design with Python experimentation and C/C++ programming. Research interests include nearest-neighbor search, clustering, optimization, and graph algorithms.

Programming & Experimental Experience

Programming: Python, C, C++.

- **Algorithmic experimentation:** Built Python simulations for nearest-neighbor search and clustering, using randomized instance generation and pattern analysis to investigate structural properties and guide theoretical work.
- **Optimization and reinforcement learning:** Evaluated convex optimization methods, including Anderson acceleration, through numerical experiments; implemented Python simulations of Thompson sampling and conducted reinforcement-learning training experiments in MuJoCo.
- **Systems and object-oriented programming:** Implemented operating-system kernel components in C/C++ and developed a World of Warcraft-themed application using object-oriented design.
- **Wireless networking:** Developed code for data analysis, simulation, and LTE base-station handover algorithms in high-speed railway networks.

Education

Max Planck Institute for Informatics | Saarland University Sept. 2022 - Present
PhD studies in Theoretical Computer Science; expected completion: April 2027.
Advisors: Danupon Nanongkai and Joachim Spoerhase.

ETH Zürich Sept. 2020 - July 2022
MSc in Computer Science; GPA: 5.62/6.0.
Major: Theoretical Computer Science; Minor: Data Management Systems.
Thesis / semester-project supervisors: Maximilian Probst Gutenberg and Rasmus Kyng.

Peking University Sept. 2016 - June 2020
BSc in Computer Science; Turing Class (inaugural cohort); GPA: 3.732/4.0.
Advisor: Zhihua Zhang.

Teaching Experience

Algorithms and Data Structures Spring 2026; Spring 2024
Teaching Assistant, Saarland University (regular course in 2026; block course in 2024).

Optimization for Data Science Spring 2022
Teaching Assistant, ETH Zürich.

Languages

Chinese (native); English (working language); German (B1); Japanese (JLPT N2).

Research Publications

FOCS 2026 | Accepted | Paper

Can We Break Fine-Grained and NP-Hardness Barriers if We've Seen the Graph Before? The Isomorphic-Priors Model

Dani Dorfman, Simon Döring, Martin Herold, Danupon Nanongkai,
Daniel Neuen, Joachim Spoerhase, and Zihang Wu.

- Introduces a model for reusing precomputed exact answers on prior graphs to accelerate computation on isomorphic inputs, without knowing the isomorphism.
- Develops exact algorithms with polynomial preprocessing and near-linear query time for constrained shortest paths, maximum flow, replacement paths, and infinite-duration games. Establishes conditional limits on this reuse for NP-complete problems.

SoCG 2025 | Paper

Sublinear Data Structures for Nearest Neighbor in Ultra High Dimensions

Martin G. Herold, Danupon Nanongkai, Joachim Spoerhase, Nithin Varma, and Zihang Wu.

- Develops approximate nearest-neighbor data structures for vectors whose dimension greatly exceeds the dataset size, answering queries without reading every coordinate of the query vector.
- Combines coordinate sampling and randomized sketching, with lower bounds showing nearly optimal dependence on the number of data points. Derives compact representations of k-median and k-means clusterings with constant-factor approximation guarantees in expectation.

SODA 2023 | Paper

Maintaining Expander Decompositions via Sparse Cuts

Yiding Hua, Rasmus Kyng, Maximilian Probst Gutenberg, and Zihang Wu.

- Develops a unified framework for maintaining expander decompositions in directed and undirected graphs under edge deletions, using local flow computations and batched updates.
- Extends the framework to expander hierarchies and derives simpler deterministic algorithms for decremental strongly connected components and fully dynamic connectivity, matching prior running times up to subpolynomial factors.

Ongoing Research

Faster Algorithms for Games on Graphs Mar. 2024 - Present

With Danupon Nanongkai, Dani Dorfman, Simon Döring, and Zhaienhe Zhou.

Investigate infinite-duration games across computational models, with the goal of developing faster sequential algorithms.

- Develop subquadratic-communication protocols for Büchi games in the two-party edge-split model, with extensions to parity games; obtain streaming space-pass tradeoffs and subquadratic cross-query bounds for Büchi games.
- Study algorithms and conditional lower bounds for game distances and dynamic attractor maintenance.